MANITHA PASANDUL

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Manitha Pasandul | Portfolio

Bandarawatta, Gampaha - 11000, Sri Lanka

Research Profile

Mechanical Engineering graduate, specialized in the Aeronautical stream from the University of Moratuwa, Sri Lanka, with First Class Honours. Experienced in flight-dynamics modelling, aerodynamic design, CFD, structural analysis, and experimental flight testing through undergraduate research, Formula Student, and industrial training. Proficient in ANSYS Fluent, MATLAB, SolidWorks, and XFLR5. As a prospective PhD student, I am interested in spacecraft and flight dynamics, aerospace propulsion, hypersonic aerodynamics, and experimental aerodynamics, particularly research combining computational modelling with experimental validation.

Education

- **University of Moratuwa** *2022 - 2026*
  BSc. (Hons) in Mechanical Engineering - Aeronautical Stream Katubedda, Sri Lanka
  - Overall GPA: 3.80/4.00 (First Class Honours)

- **Bandaranayake College** *2012 - 2020*
  GCE Advanced Level Examination, Physical Science Stream Gampaha, Sri Lanka
  - Z-score: 2.1187
  - GCE Ordinary Level Examination (2017): 9 As

- **Sri Bodhi Primary College** *2007 - 2011*
  Primary Education Gampaha, Sri Lanka

Work Experience

- **Temporary Instructor** *July 2026 – Present*
  Department of Mechanical Engineering, University of Moratuwa, Moratuwa, Sri Lanka
  - Support the delivery of tutorials, practical sessions, and laboratory activities for Manufacturing Processes, Basic Thermodynamics, Mechanics of Machines I, and Principles of Naval Architecture
  - Assist academic staff with laboratory preparation, practical demonstrations, student supervision, and the coordination of course-related activities

- **Trainee Aeronautical Engineer** *Jan 2025 – Jun 2025*
  SriLankan Airlines, Bandaranaike International Airport, Katunayake, Sri Lanka
  - Gained hands-on exposure through workshops on engine modules, wheels and brakes, NDT, paint and composite repairs, and structural maintenance
  - Developed practical understanding of aircraft sub-systems by observing and assisting inspection, disassembly, repair, and reassembly processes

- **Aerodynamics Engineer** *Dec 2025 – Present*
  FalconE Racing Team
  - Designed, optimized and fabricated aerodynamic components using CFD analysis to improve downforce, drag performance, and manufacturability
  - Supported structural integration and carbon fiber composite fabrication of aerodynamic components for the Formula Student race car

PROJECTS

- **Flight Dynamics Modelling and Validation of Small Fixed-Wing UAVs** *Aug 2025 – June 2026*

  Tools: Python, AVL, FLOW5, XFOIL, MATLAB, SolidWorks

  - Developed a Python-based framework for fixed-wing UAV flight-dynamics analysis, using reduced longitudinal and lateral-directional models derived from the aircraft equations of motion around trimmed flight conditions.
  - Estimated aerodynamic and stability derivatives using Vortex Lattice Method (VLM) simulations and performed eigenvalue analysis to evaluate natural frequencies, damping ratios, and time constants of short-period, phugoid, Dutch-roll, roll-subsidence, and spiral modes.
  - Conducted parametric sensitivity analyses of aircraft geometry, including wing dihedral, and tail volume ratios, to investigate their effects on dynamic stability and develop sensitivity-guided flying-quality tuning strategies.
  - Designed and fabricated two fixed-wing UAVs, including a 2.5 m wingspan platform, and conducted flight tests using controlled elevator disturbances with onboard IMU and control-input data logging.
  - Processed flight-test measurements using Fast Fourier Transform (FFT) and Prony analysis to identify experimental modal frequencies and damping characteristics, comparing the results with theoretical predictions to evaluate model accuracy.
  - Co-authored the research paper *"A Python-Based Framework for Conventional Surveillance UAV Design and Sensitivity-Guided Flying-Quality Tuning,"* currently under review for the iMERS conference in Sri Lanka.

- **Aircraft Propeller Design and Performance Analysis** *Jun 2025 – Dec 2025*

  Tools: MATLAB, SolidWorks

  - Designed a two-bladed aircraft propeller using blade element theory and iterative MATLAB calculations
  - Determined blade geometry and estimated thrust, torque, inflow factors, efficiency, and required engine power

- **CFD Analysis of NREL S809 Aerofoil** *June 2025 – Dec 2025*

  Tools: ANSYS Fluent, MATLAB

  - Performed 2D steady-state CFD analysis of the NREL S809 aerofoil for wind turbine blade applications
  - Evaluated lift, drag, lift-to-drag ratio, pressure contours, and velocity contours for angles of attack from 0° to 20°

- **Experimental and Numerical Stress Analysis of a Wing Structure** *Jan 2026 – June 2026*

  Tools: ANSYS Static Structural

  - Developed an ANSYS finite element model to predict stress and strain distributions under static loading
  - Validated FEA results using experimental strain gauge measurements and identified possible sources of deviation

- **Design of a Sailplane Wing** *Aug 2024 – Dec 2024*

  Tools: XFLR5, MATLAB, SolidWorks

  - Designed a high-performance sailplane wing through aerofoil selection, finite-wing analysis, and glider performance calculations
  - Designed a dual-spar tapered wing structure and evaluated stress, deflection, shear flow, buckling, and material selection

- **Design and Development of Front Wing Aerodynamic Package** *Jan 2025 – Dec 2025*

  Tools: Ansys Fluent, SolidWorks

  - Designed a four-element 2D front wing profile and optimized its aerodynamic geometry, using Multi-Objective Genetic Algorithm (MOGA) to improve downforce while maintaining low drag profile
  - Developed structural components and fabricated the final wing using carbon fiber composite layup techniques for lightweight and high-stiffness performance

Honors and Awards

- **Dean's List - Semester 2** *July 2023*
  University of Moratuwa | Semester GPA: 3.86
- **Dean's List - Semester 4** *Aug 2024*
  University of Moratuwa | Semester GPA: 3.91
- **Dean's List - Semester 6** *Aug 2025*
  University of Moratuwa | Semester GPA: 4.00
- **Dean's List - Semester 7** *Dec 2025*
  University of Moratuwa | Semester GPA: 3.85
- **Dean's List - Semester 8** *July 2026*
  University of Moratuwa | Semester GPA: 4.00

Volunteer Experience

- **Volunteer Teacher (Ganitha Sawiya)** *2022 – 2023*
  Sasnaka Sansadaya
- **Volunteer - Aerodynamics Team Member** *Aug 2024 – Dec 2025*
  FalconE Racing Team
- **Demonstrator at Techno 2024** *Oct 2024*
  FalconE Racing Team

Professional Memberships

- **Student Member** Institution of Engineers Sri Lanka (IESL) *Mar 2024 - Present*
  Membership No: S-30208
- **Student Affiliate Member** Institution of Mechanical Engineers (IMechE) *Dec 2022 - Present*
  Membership No: 80652300

Additional Information

- **IELTS Academic:** Overall 7.5 (L: 8.5, R: 8.0, W: 7.0, S: 6.5) – September 2026
- **Languages:** English (Proficient), Sinhala (Native)
- **Interests:** Space Systems, Hypersonic Aerodynamics, Aerospace Propulsion, Flight Dynamics and Control

References

1. **Dr. Nalaka Samaraweera**
   Senior Lecturer, Department of Mechanical Engineering
   University of Moratuwa
   Email: nalakas@uom.lk
   Relation: Project Supervisor
2. **Prof. Nirosh Jayaweera**
   Head of the Department, Department of Mechanical Engineering
   University of Moratuwa
   Email: niroshj@uom.lk
   Relation: Project Supervisor